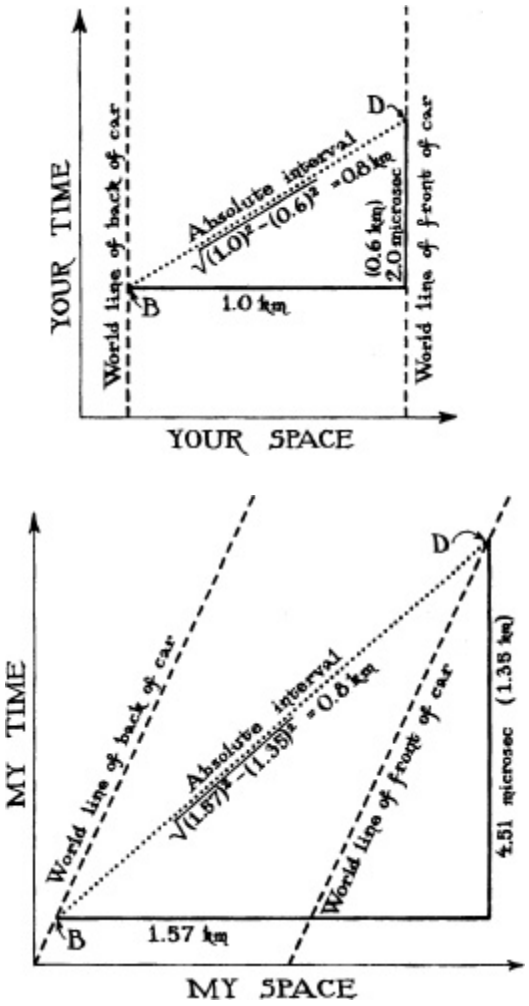


Because space and time are relative (your space is a mixture of my space and my time), you and I disagree about the time separation between the backfire event *B* and the detonation event *D.* They are separated by 2.0 microseconds of your time, and by 4.51 microseconds of mine. Similarly, we disagree about the events' spatial separation; it is 1.0 kilometer in your space and 1.57 kilometers in mine. Despite these temporal and spatial disagreements, we *agree* that the two events are separated by a straight line in four-dimensional spacetime, and we *agree* that the "absolute interval" along that line (the spacetime length of the line) is 0.8 kilometer. (This is analogous to the Mledinan men and women agreeing on the straight-line distance between Mledina and Serona.)

We can use Minkowski's formula to compute the absolute interval: We each multiply the events' time separation by the speed of light (299,792 kilometers per second), getting the rounded-off numbers shown in the diagrams (0.600 kilometer for you, 1.35 kilometers for me). We then square